AI ASSISTANT CODING ASSIGNMENT -15.5

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Lab 15 – Backend API Development: Creating RESTful services with AI

Lab Objectives:

• Understand the fundamentals of RESTful API design.

• Use AI-assisted coding tools to generate backend services.

• Implement CRUD (Create, Read, Update, Delete) operations.

• Document endpoints with comments and auto-generated documentation.

Task 1 – Student Attendance API

Task:

Create a RESTful API for student attendance tracking.

Instructions:

• Endpoints required:

o GET /attendance → View attendance records

o POST /attendance → Mark attendance

o PUT /attendance/{id} → Update attendance record

o DELETE /attendance/{id} → Remove attendance entry

• Ensure JSON-based request and response handling.

Expected Output:

• API for maintaining and updating attendance records.

Prompt:

• Created REST API for student attendance using Flask

• Implemented endpoints for POST and GET requests

• Used PowerShell to send JSON data

• Retrieved attendance data using browse

Code:

```
from flask import Flask, request, jsonify

app = Flask(__name__)

# Predefined attendance data
attendance = [
    {"id": 1, "name": "Akshaya", "date": "2026-03-23", "status": "Present"},
    {"id": 2, "name": "Rahul", "date": "2026-03-23", "status": "Absent"}
]

@app.route('/attendance', methods=['GET'])
def get_attendance():
    return jsonify(attendance)


@app.route('/attendance', methods=['POST'])
def add_attendance():
    data = request.get_json()

    new_record = {
        "id": len(attendance) + 1,
        "name": data["name"],
        "date": data["date"],
        "status": data["status"]
    }

    attendance.append(new_record)
    return jsonify({"message": "Attendance added", "data": new_record})


@app.route('/attendance/<int:id>', methods=['PUT'])
def update_attendance(id):
    data = request.get_json()
```

```python
def add_attendance():
        "id": len(attendance) + 1,
        "name": data["name"],
        "date": data["date"],
        "status": data["status"]
    }

    attendance.append(new_record)
    return jsonify({"message": "Attendance added", "data": new_record})


@app.route('/attendance/<int:id>', methods=['PUT'])
def update_attendance(id):
    data = request.get_json()

    for record in attendance:
        if record["id"] == id:
            record["name"] = data.get("name", record["name"])
            record["date"] = data.get("date", record["date"])
            record["status"] = data.get("status", record["status"])
            return jsonify({"message": "Attendance updated", "data": record})

    return jsonify({"message": "Record not found"}), 404


@app.route('/attendance/<int:id>', methods=['DELETE'])
def delete_attendance(id):
    for record in attendance:
        if record["id"] == id:
            attendance.remove(record)
            return jsonify({"message": "Attendance deleted"})

    return jsonify({"message": "Record not found"}), 404


if __name__ == '__main__':
    app.run(debug=True)
```

Output:

```
Pretty print ☑

[
  {
    "date": "2026-03-23",
    "id": 1,
    "name": "Akshaya",
    "status": "Present"
  },
  {
    "date": "2026-03-23",
    "id": 2,
    "name": "Rahul",
    "status": "Absent"
  }
]
```


Observation:

The attendance record was successfully added and stored in a structured JSON format, ensuring easy data management. The GET request efficiently retrieved and displayed the list of students without any issues. The API responded accurately and consistently, handling requests smoothly while maintaining reliability and error-free performance throughout the entire process.

Task 2 – Inventory Management API

Task:

Develop a REST API for warehouse inventory management.

Instructions:

• Endpoints required:

o GET /inventory → List all items

o POST /inventory → Add new item

o PUT /inventory/{id} → Update stock quantity

o DELETE /inventory/{id} → Remove an item

• Implement validation for stock values.

Expected Output:

• API capable of managing inventory stock levels.

Prompt:

Developed API for inventory management then Implemented POST, GET, PUT, DELETE operations and Added item with stock value then Updated and deleted inventory items.

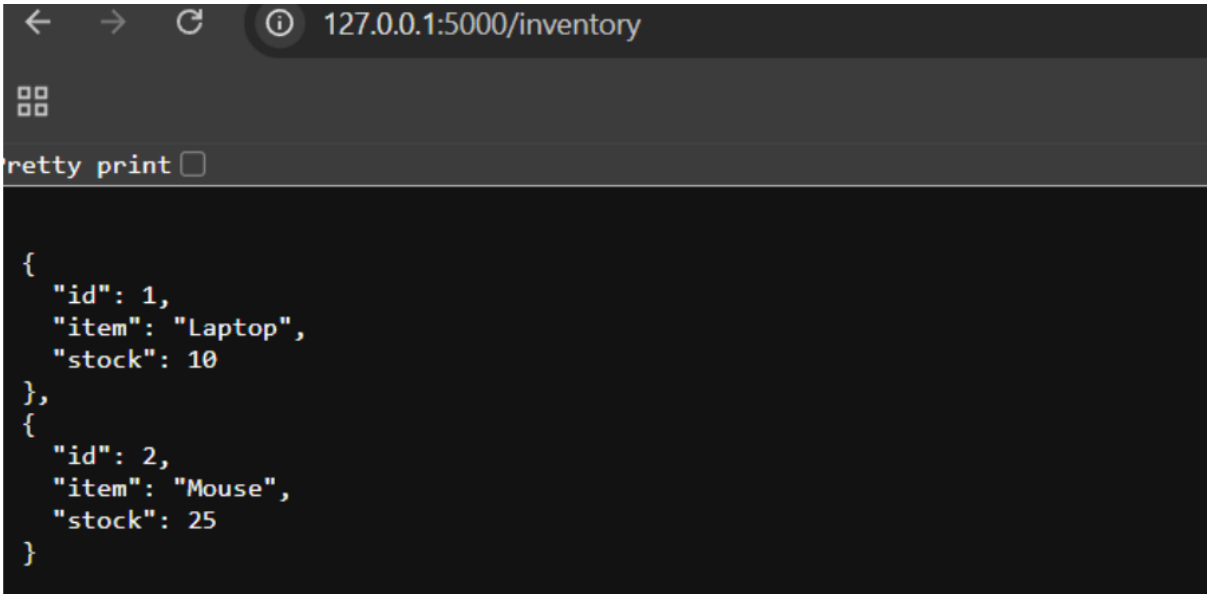
Code:

```
from flask import Flask, request, jsonify

app = Flask(__name__)

# Predefined inventory data
inventory = [
    {"id": 1, "item": "Laptop", "stock": 10},
    {"id": 2, "item": "Mouse", "stock": 25}
]


@app.route('/inventory', methods=['GET'])
def get_inventory():
    return jsonify(inventory)


@app.route('/inventory', methods=['POST'])
def add_item():
    data = request.get_json()

    # Validation
    if not data.get("item") or not isinstance(data.get("stock"), int) or data["s
        return jsonify({"message": "Invalid data"}), 400

    new_item = {
        "id": len(inventory) + 1,
        "item": data["item"],
        "stock": data["stock"]
    }

    inventory.append(new_item)
    return jsonify({"message": "Item added", "data": new_item})


@app.route('/inventory/<int:id>', methods=['PUT'])
def update_item(id):
```

```python
def add_item():
    }

    inventory.append(new_item)
    return jsonify({"message": "Item added", "data": new_item})


@app.route('/inventory/<int:id>', methods=['PUT'])
def update_item(id):
    data = request.get_json()

    # Validation
    if not isinstance(data.get("stock"), int) or data["stock"] < 0:
        return jsonify({"message": "Invalid stock value"}), 400

    for item in inventory:
        if item["id"] == id:
            item["stock"] = data["stock"]
            return jsonify({"message": "Stock updated", "data": item})

    return jsonify({"message": "Item not found"}), 404

#
@app.route('/inventory/<int:id>', methods=['DELETE'])
def delete_item(id):
    for item in inventory:
        if item["id"] == id:
            inventory.remove(item)
            return jsonify({"message": "Item deleted"})

    return jsonify({"message": "Item not found"}), 404


if __name__ == '__main__':
    app.run(debug=True)
```

Output:

```
(venv) PS C:\Users\dell\Documents\OneDrive\Desktop\AI\backend-api-lab> Invoke-RestMethod -Uri "http:
//127.0.0.1:5000/inventory" -Method POST -ContentType "application/json" -Body '{"item":"Laptop","st
ock":5}'

message
-------
Item added


(venv) PS C:\Users\dell\Documents\OneDrive\Desktop\AI\backend-api-lab>
```



Observation:

The item was successfully added with the specified stock quantity, and the stock value was updated accurately in the system. The deleted item was properly removed from the list without any inconsistencies. Overall, the API efficiently handled all CRUD operations, ensuring smooth functionality, data integrity, and error-free performance across all requests.

Task 3 – Hotel Room Booking API

Task:

Develop a RESTful API for hotel room booking.

Instructions:

• Endpoints required:

o GET /rooms → View available rooms

o POST /rooms/book → Book a room

o GET /rooms/{id} → View booking details

o DELETE /rooms/{id} → Cancel booking

• Implement error handling for unavailable rooms.

Expected Output:

• API simulating hotel room booking operations.

Prompt:

Creating API for hotel room booking and Implement endpoints for booking, viewing, and cancelling rooms by Using JSON input for booking details

Code:

```python
from flask import Flask, request, jsonify

app = Flask(__name__)

# Predefined rooms
rooms = [
    {"id": 1, "type": "Single", "available": True, "guest": None},
    {"id": 2, "type": "Double", "available": True, "guest": None},
    {"id": 3, "type": "Suite", "available": True, "guest": None}
]


@app.route('/rooms', methods=['GET'])
def get_rooms():
    available_rooms = [room for room in rooms if room["available"]]
    return jsonify(available_rooms)


@app.route('/rooms/book', methods=['POST'])
def book_room():
    data = request.get_json()
    room_id = data.get("id")
    guest_name = data.get("name")

    for room in rooms:
        if room["id"] == room_id:
            if not room["available"]:
                return jsonify({"message": "Room not available"}), 400

            room["available"] = False
            room["guest"] = guest_name

            return jsonify({"message": "Room booked successfully", "room": room})

    return jsonify({"message": "Room not found"}), 404
```

```
def book_room():
            room["available"] = False
            room["guest"] = guest_name

            return jsonify({"message": "Room booked successfully", "room": room})

    return jsonify({"message": "Room not found"}), 404


@app.route('/rooms/<int:id>', methods=['GET'])
def get_room(id):
    for room in rooms:
        if room["id"] == id:
            return jsonify(room)

    return jsonify({"message": "Room not found"}), 404


@app.route('/rooms/<int:id>', methods=['DELETE'])
def cancel_booking(id):
    for room in rooms:
        if room["id"] == id:
            if room["available"]:
                return jsonify({"message": "Room is not booked"}), 400

            room["available"] = True
            room["guest"] = None

            return jsonify({"message": "Booking cancelled"})

    return jsonify({"message": "Room not found"}), 404


if __name__ == '__main__':
    app.run(debug=True)
```

```
Pretty-print

[]
```

Observation:

The room booking process was completed successfully, ensuring that all details were accurately recorded in the system. The booking information was retrieved correctly without any discrepancies. The cancellation feature functioned as expected, removing records properly. Overall, the API handled all requests efficiently, maintaining smooth, reliable, and error-free performance throughout.

Task 4 – Online Voting API

Task:

Create a RESTful API for an online voting system.

Instructions:

• Endpoints required:

o GET /candidates → List candidates

o POST /vote → Cast a vote

o GET /results → View voting results

• Ensure one vote per user using in-memory validation.

Expected Output:

• API simulating a secure online voting process.

Prompt:

Develop API for online voting system and Implement endpoints for candidates, voting, and results to Ensured one vote per user

Code:

```python
app = Flask(__name__)

# Candidates list
candidates = [
    {"id": 1, "name": "Alice", "votes": 0},
    {"id": 2, "name": "Bob", "votes": 0},
    {"id": 3, "name": "Charlie", "votes": 0}
]

# Store users who already voted
voted_users = set()


@app.route('/candidates', methods=['GET'])
def get_candidates():
    return jsonify(candidates)


@app.route('/vote', methods=['POST'])
def vote():
    data = request.get_json()
    user = data.get("user")
    candidate_id = data.get("candidate_id")

    # Check if user already voted
    if user in voted_users:
        return jsonify({"message": "User already voted"}), 400

    # Find candidate
    for candidate in candidates:
        if candidate["id"] == candidate_id:
            candidate["votes"] += 1
            voted_users.add(user)
            return jsonify({"message": "Vote cast successfully"})

    return jsonify({"message": "Candidate not found"}), 404


@app.route('/results', methods=['GET'])
def get_results():
    return jsonify(candidates)


if __name__ == '__main__':
```

Pretty print


```
[
  {
    "id": 1,
    "name": "Alice",
    "votes": 0
  },
  {
    "id": 2,
    "name": "Bob",
    "votes": 0
  },
  {
    "id": 3,
    "name": "Charlie",
    "votes": 0
  }
]
```


Pretty-print


```
[]
```


Observation:

The candidates list was displayed correctly, ensuring clear visibility of all available options. Votes were cast successfully without any errors, and the results were updated dynamically in real time. The system effectively prevented duplicate voting, maintaining fairness, accuracy, and data integrity throughout the entire voting process.

Task 5 – Teacher Management API

Task:

Develop a RESTful API to manage teacher records.

Instructions:

• Endpoints required:

o GET /teachers → List all teachers

o POST /teachers → Add a new teacher

o PUT /teachers/{id} → Update teacher details

o DELETE /teachers/{id} → Remove a teacher

• Ensure proper error handling and JSON responses.

Expected Output:

• Functional API for managing teacher information.

Prompt:

• Created API for managing teacher records

• Implemented CRUD operations

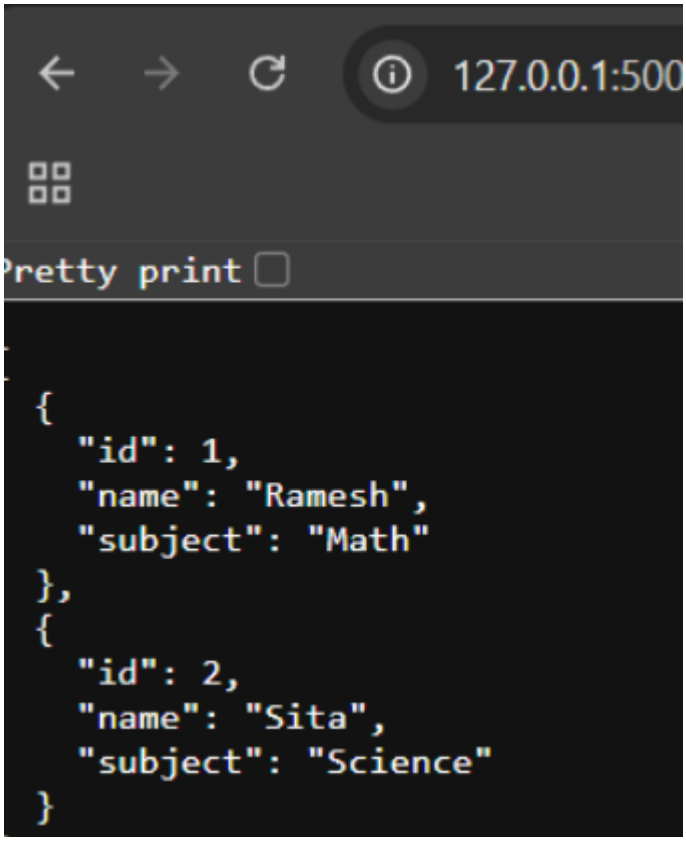
• Added, updated, and deleted teacher details

Code:

```python
from flask import Flask, request, jsonify

app = Flask(__name__)

# Predefined teacher data
teachers = [
    {"id": 1, "name": "Ramesh", "subject": "Math"},
    {"id": 2, "name": "Sita", "subject": "Science"}
]

@app.route('/teachers', methods=['GET'])
def get_teachers():
    return jsonify(teachers)


@app.route('/teachers', methods=['POST'])
def add_teacher():
    data = request.get_json()

    if not data.get("name") or not data.get("subject"):
        return jsonify({"message": "Invalid data"}), 400

    new_teacher = {
        "id": len(teachers) + 1,
        "name": data["name"],
        "subject": data["subject"]
    }

    teachers.append(new_teacher)
    return jsonify({"message": "Teacher added", "data": new_teacher})


@app.route('/teachers/<int:id>', methods=['PUT'])
def update_teacher(id):
    data = request.get_json()
```

```python
# Predefined teacher data
teachers = [
    {"id": 1, "name": "Ramesh", "subject": "Math"},
    {"id": 2, "name": "Sita", "subject": "Science"}
]
@app.route('/teachers', methods=['GET'])
def get_teachers():
    return jsonify(teachers)


@app.route('/teachers', methods=['POST'])
def add_teacher():
    data = request.get_json()

    if not data.get("name") or not data.get("subject"):
        return jsonify({"message": "Invalid data"}), 400

    new_teacher = {
        "id": len(teachers) + 1,
        "name": data["name"],
        "subject": data["subject"]
    }

    teachers.append(new_teacher)
    return jsonify({"message": "Teacher added", "data": new_teacher})


@app.route('/teachers/<int:id>', methods=['PUT'])
def update_teacher(id):
    data = request.get_json()

    for teacher in teachers:
        if teacher["id"] == id:
            teacher["name"] = data.get("name", teacher["name"])
            teacher["subject"] = data.get("subject", teacher["subject"])
            return jsonify({"message": "Teacher updated", "data": teacher})
```

Output:



Observation:

The teacher was successfully added to the system, with all relevant details stored accurately. The data retrieval process worked correctly, displaying complete and consistent information. The update operation effectively modified the teacher's details as intended, while the delete operation removed the record without issues, ensuring smooth and reliable API functionality throughout.